4th Down Dilemma: An NFL Fourth Down Conversion Rate Analysis

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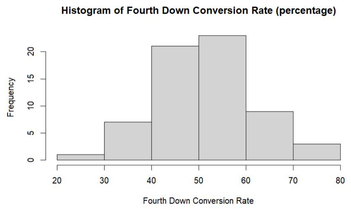
Background

- Deciding whether to punt, attempt a field goal, or go for it is one of the most difficult decisions in football
- Statistics show that the average NFL team's success rate at converting fourth down is 50%, but NFL teams only go for it 20% of the time (Cavanagh, 2023)
- NFL is leaning more into analytics (Sutelan, 2023)

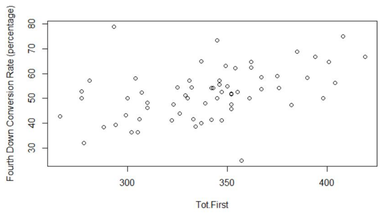
Research Questions:

1. How does the total number of first downs that different NFL teams get effect 4th down conversion percentage in the regular season?
2. How many times do different NFL teams typically attempt 4th down conversions in a season, and is this effective at predicting their 4th down conversion percentage in the regular season?
3. What is the number of yards gained by different NFL teams on 4th down conversion attempts in the regular season, and does it correlate with 4th down conversion percentage?

Exploratory Data Analysis



Fourth down conversion rate has a relatively normal distribution, so we knew that it would be compatible for multiple linear regression.



	[,1]
Tot.First	0.442
Pct.Third	0.273
Att.Fourth	-0.025
Avg.Yards.Fourth	0.185
Seconds.Rem	0.341
Yards.To.Go	-0.312
Yards.To.TD	0.006
Yds.per.Game	0.362

None of our quantitative predictive variables had a very strong correlation with the response variable, but total first downs had the highest correlation value of 0.442. It ended up being one of two quantitative predictors in our model.

Data Summary

Data was collected from the following sources:

- Football Database
- Pro Football Focus (PFF)
- NFL Verse package in R
- Data collected for all 32 teams during the 2021 and 2022 NFL regular seasons for a total of 64 observations.
- Football Database - 4 quantitative explanatory variables and a 5th quantitative response variable (4th down conversion percentage).
- PFF - offensive line rankings converted into boolean variables recording a 1 if o-line performed in the top half of the league and a 0 if they didn't.
- NFL Verse - Since the package was on a play-by-play basis, we manipulated 4 quantitative explanatory play-by-play variables to retrieve season long average values for each observation.

Multiple Linear Regression Analysis

Step 1: Quantitative Variables

Multicollinearity

	Tot.First	Pct.Third	Att.Fourth	Avg.Yards.Fourth	Seconds.Rem	Yards.To.Go	Yards.To.TD
Tot.First	1.00	0.70	-0.02	0.16	0.47	-0.52	-0.36
Pct.Third	0.70	1.00	-0.35	-0.05	0.40	-0.43	-0.44
Att.Fourth	-0.02	-0.35	1.00	0.17	0.06	0.03	0.09
Avg.Yards.Fourth	0.16	-0.05	0.17	1.00	-0.07	0.26	0.40
Seconds.Rem	0.47	0.40	0.06	-0.07	1.00	-0.44	-0.20
Yards.To.Go	-0.52	-0.43	0.03	0.26	-0.44	1.00	0.37
Yards.To.TD	-0.36	-0.44	0.09	0.40	-0.20	0.37	1.00
Yds.per.Game	0.91	0.69	-0.16	0.12	0.45	-0.53	-0.44

	Yds.per.Game
Tot.First	0.91
Pct.Third	0.69
Att.Fourth	-0.16
Avg.Yards.Fourth	0.12
Seconds.Rem	0.45
Yards.To.Go	-0.53
Yards.To.TD	-0.44
Yds.per.Game	1.00

Multiple pairwise relationships have high collinearity values, indicating there may be high multicollinearity.

Original Equation: Pct.Fourth = β0 + β1Tot.First + β2Pct.Third + β3Att.Fourth + β4Avg.Yards.Fourth + β5Seconds.Rem + β6Yards.To.Go + β7Yards.To.TD + β8Yds.per.Game

Variable Screening

```{r, echo=FALSE, warning=FALSE,results=FALSE}
ols_step_both_p(fourthdownmod1,pent=0.15,prem=0.15,details=T)
```

Stepwise Selection Summary

Step	Variable	Added/ Removed	R-Square	Adj. R-Square	C(p)	AIC	RMSE
1	Tot.First	addition	0.195	0.182	2.0340	472.5108	9.4079
2	Yards.To.TD	addition	0.226	0.201	1.6430	471.9946	9.3001

New Equation: Pct.Fourth = □1Tot.First + □2Yards.To.TD

Step 2: Qualitative Variables

Original Equation: Pct.Fourth = β1Tot.First + β2Yards.To.TD + β3Conference + β4Playoffs + β5Top.Oline

New Equation: Pct.Fourth = β1Tot.First + β2Yards.To.TD + β3Conference

Step 3: Interactions

Original Equation: Pct.Fourth = β1Tot.First + β2Yards.To.TD + β3Tot.First*Yards.To.TD + β4Conference

Final Equation: Pct.Fourth = β1Tot.First + β2Yards.To.TD + β3Tot.First*Yards.To.TD + β4Conference

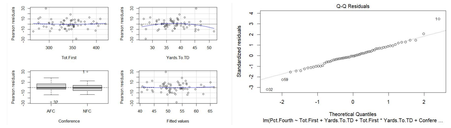
```
Call:
lm(formula = Pct.Fourth ~ Tot.First + Yards.To.TD + Tot.First *
    Yards.To.TD + Conference, data = fourthdowndata)

Residuals:
     Min       1Q   Median       3Q      Max
-27.9615  -4.8777  -0.4544   5.2498  29.7116

Coefficients:
                          Estimate Std. Error t value Pr(>|t|)
(Intercept)             -1.207e+02  7.357e+01  -1.641   0.1062
Tot.First                4.630e-01  2.102e-01   2.203   0.0315 *
Yards.To.TD              3.239e+00  1.892e+00   1.712   0.0922 .
ConferenceNFC           -3.451e+00  2.297e+00  -1.502   0.1383
Tot.First:Yards.To.TD -8.223e-03  5.466e-03  -1.505   0.1378
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

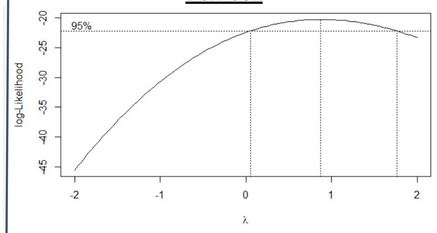
Residual standard error: 9.106 on 59 degrees of freedom
Multiple R-squared:  0.2824,    Adjusted R-squared:  0.2337
F-statistic: 5.804 on 4 and 59 DF,  p-value: 0.0005234
```

Step 4: Regression Assumptions



Since there does not appear to a mean of zero at all points in each residual plot, the mean zero assumption is violated. Residuals have a further distance from the mean at some points than at other points in the residual plots and residuals vs fitted plot, so the constant variance assumption is violated. Since the QQ plot is skewed at its ends, the normality assumption is violated. Since we do not have time series data, the independence assumption has not been violated.

Box-Cox



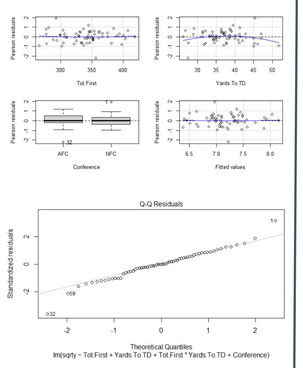
Lambda's value is a little below 1, indicating that transforming the response variable using a square root would be ideal.

```
Call:
lm(formula = sqrty ~ Tot.First + Yards.To.TD + Tot.First * Yards.To.TD +
    Conference, data = fourthdowndata)

Residuals:
     Min       1Q   Median       3Q      Max
-2.22014 -0.32472  0.00005  0.37192  1.90755

Coefficients:
                          Estimate Std. Error t value Pr(>|t|)
(Intercept)             -4.257041   5.182684  -0.821   0.4147
Tot.First                0.030339   0.014806   2.049   0.0449 *
Yards.To.TD              0.209065   0.133304   1.568   0.1222
ConferenceNFC           -0.232925   0.161817  -1.439   0.1553
Tot.First:Yards.To.TD -0.000520   0.000385  -1.350   0.1820
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6415 on 59 degrees of freedom
Multiple R-squared:  0.2761,    Adjusted R-squared:  0.227
F-statistic: 5.625 on 4 and 59 DF,  p-value: 0.0006638
```

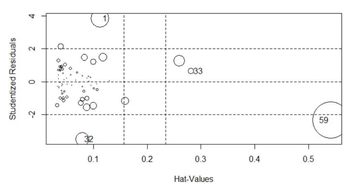


This also violates the assumptions of zero mean, constant variance, and normality.

Multiple Linear Regression Analysis

Step 5: Influential Observation Analysis

	StudRes	Hat	CookD
1	3.8466605	0.11225440	0.30328597
32	-3.4889648	0.07872508	0.17491609
33	0.6473172	0.28205785	0.03325147
59	-2.3449042	0.54159098	1.20722022



Observations 1, 32, 33, and 59 are outliers. Since transforming the response variable did not prevent violations of regression assumptions, we removed these observations.

```{r}
subsetfourthdowndata<-fourthdowndata[-c(1,32,33,59),]
fourthdownsubmod<-lm(Pct.Fourth~Tot.First+Yards.To.TD+Tot.First*Yards.To.TD+Conference, data = subsetfourthdowndata)
summary(fourthdownsubmod)
```

```
Call:
lm(formula = Pct.Fourth ~ Tot.First + Yards.To.TD + Tot.First *
    Yards.To.TD + Conference, data = subsetfourthdowndata)

Residuals:
     Min       1Q   Median       3Q      Max
-15.6417  -4.4054   0.0543   4.8204  18.2511

Coefficients:
                          Estimate Std. Error t value Pr(>|t|)
(Intercept)             -1.864e+02  8.697e+01  -2.143  0.03653 *
Tot.First                6.791e-01  2.463e-01   2.758  0.00789 **
Yards.To.TD              5.091e+00  2.351e+00   2.166  0.03469 *
ConferenceNFC           -5.314e+00  1.881e+00  -2.825  0.00657 **
Tot.First:Yards.To.TD -1.423e-02  6.704e-03  -2.123  0.03828 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 7.16 on 55 degrees of freedom
Multiple R-squared:  0.4434,    Adjusted R-squared:  0.4029
F-statistic: 10.95 on 4 and 55 DF,  p-value: 1.325e-06
```

This is our final model after subsetting.

Variable	Variable Type	Variable Meaning
Pct.Fourth	Response, Quantitative	Percentage of Fourth Downs Converted
Tot.First	Explanatory, Quantitative	Total First downs converted by team
Yards.To.TD	Explanatory, Quantitative	Yards left to score a touchdown
ConferenceNFC	Explanatory, Qualitative	If the team is in the Conference NFC

Conclusions

Final Prediction Equation:

$$Pct.Fourth = -.01864 + .6791(Tot.First) + 5.091(Yards.To.TD) - 5.314(ConferenceNFC) - 1.423(Tot.First * Yards.To.TD)$$

- Through building different models and analyzing the results, we decided on choosing this final prediction equation. The equation is predicting the fourth down percentage conversion, by using our explanatory variables. For the final equation, it contains total first downs and yards to touchdown, our two quantitative variables. It also contains one of our qualitative variables, Conference. Finally, there is an interaction term between our two quantitative variables in the equation. All of the p-values for the variables are below .05, and the p-value for the f-test is 1.325*10^-6. This tells us that not only are all of our predictor variables significant, but also the model itself. However, our adjusted R-squared value of .403 is not ideal.
- Total number of first downs is a telling predictor of how good of an offense a team has. It comes as no surprise that this is our most significant predictor of fourth down conversion percentage. In the model, each first down that a team accumulates throughout the season adds .6791 percent to their fourth down conversion percentage in the context of our final prediction equation.

Future Research

- For this project, the only offensive players that we analyzed was the offensive line. For future works, we could look into different offensive positions and their effects on fourth down conversion ratings to see if any of them are significant.
- Due to a the league wide adaptation of the Philadelphia Eagles play called the "The Tush Push" the 4th and short conversion rates for the 2023 season are higher than previous years. Since the 2023 season is not included in our dataset, our model will be weaker at prediction. Once the 2023 regular season is finished, we would be able to incorporate it into our model and see how it changes it.
- Our model only includes regular season data, although the model considers whether a team made the playoffs. In the future, we could make a model that incorporates playoff data or is for teams in the playoffs specifically.

Works Cited

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